Data Quality Report

Overview

Description of the Analytics Base Table (ABT)

The training CSV file represents the initial Analytics Base Table (ABT) used for the property price prediction problem. Each row corresponds to a single property transaction, and the goal of the predictive model is to estimate the sale price of a property based on the available descriptive features.

The key characteristics of the dataset are summarised below:

Attribute	Description
Number of rows (before cleaning)	54,000
Number of features	9
Target variable	Price (€)
Unit of analysis	Property transaction
Time coverage	02/2016 - 05/2024

Aim of the Data Quality Report

The aim of this report is to evaluate the quality of the dataset by analysing structure, completeness, and statistical properties of features, and to identify issues that may impact predictive modelling.

ABT Structure & Feature Typing

Feature Classification

The table below shows how we have classified each feature and our reasoning for doing so.

Feature	Type	Rationale
Price	Continuous (Numeric)	Feature contains numeric values that allow arithmetic operations.
County	Categorical (Nominal)	There is a finite set of values that cannot be ordered and allow no arithmetic.
Not Full Market Price	Categorical (Binary)	There are two values that cannot be ordered and do not allow arithmetic.
VAT Exclusive	Categorical (Binary)	There are two values that cannot be ordered and do not

		allow arithmetic.
Description of Property	Categorical (Nominal)	There is a finite set of values that cannot be ordered and allow no arithmetic.
Property Size Description	Categorical (Ordinal)	Values allow ordering but do not permit arithmetic.
Eircode	Categorical (Nominal, with very high cardinality)	There is a finite set of values that cannot be ordered and allow no arithmetic.
Address	Textual	Free form, short, text data.
Date of Sale	Interval/Temporal	Values that allow ordering and subtraction, but do not allow other arithmetic operations.

(Kelleher, Mac Namee, D'Arcy; 2020)

Continuous Features: Data Quality Assessment

Table A.1 in Appendix A presents the Data Quality Report (DQR) summary statistics for the continuous features in the dataset. The table includes the count of observations, percentage of missing values, cardinality, and measures of central tendency and variation including the mean, median, quartiles, minimum, maximum and standard deviation. These statistics provide an overview of the distribution and spread of values for the continuous variables.

The standard deviation is approximately €1,110,000, which is considerably larger than the mean. This indicates very high variability in property prices across the dataset.

The minimum recorded sale price is €5,260 while the maximum price is €136,000,000. This extremely wide range further confirms that the distribution of prices contains substantial variation and several extreme values.

Initial visualisation using a histogram (Appendix A.2) did not provide a clear view of the distribution due to the presence of extremely large values stretching the x-axis. A boxplot (Appendix A.3) revealed the presence of numerous high-value outliers. To better visualise the underlying distribution, a second histogram limited to the 99th percentile was produced (Appendix A.4). (The data has not been clamped yet, this is just a brief visualisation.) This histogram shows a unimodal distribution that is strongly right-skewed, with most transactions clustered in the lower price ranges and a small number of very high-value sales forming a long upper tail. The large difference between the mean and median price further confirms that the distribution is strongly right-skewed.

Categorical Features: Data Quality Assessment

Table B.1 in Appendix C presents the Data Quality Report (DQR) summary statistics for the categorical features in the dataset. For each feature, the table reports the number of observations, percentage of missing values, and the number of distinct levels (cardinality). It also identifies the two most frequent categories (mode and second mode), along with their frequencies and proportions within the dataset. These statistics provide insight into the distribution of categorical variables and help identify dominant categories, high-cardinality features, and potential data quality issues.

The **County** feature contains 26 distinct levels and no missing values, indicating complete geographic information for all observations. Dublin is the most frequent category, representing approximately 30.7% of transactions, followed by Cork with approximately 11.2%. The distribution of counties suggests that property transactions are concentrated in a small number of regions, particularly Dublin. This distribution can be seen in the bar plot presented in Appendix B.2.

The **Not Full Market Price** feature is a binary categorical variable with two levels: “Yes” and “No”. The majority of transactions (approximately 94.7%) are recorded as full market price sales, while only 5.28% are marked as not full market price. The strong dominance of the “No” category suggests that most recorded property transactions represent market-value sales. The distribution of this feature is illustrated in Appendix B.3.

The **VAT Exclusive** feature is also binary with two possible values. Approximately 82.5% of transactions are recorded as not VAT exclusive, while 17.5% are VAT exclusive. This suggests that the majority of property sales include VAT within the recorded transaction price. The bar plot for this feature is shown in Appendix B.4. An additional observation is that the VAT Exclusive feature is highly correlated with the Description of Property feature. Specifically, VAT-exclusive transactions almost always correspond to new dwellings, with approximately 99.7% agreement between the two variables. A small number of inconsistencies are present, indicating that while the features are closely related, they are not perfectly identical.

The **Description of Property** feature contains three distinct categories and no missing values. The most common category is “Second-Hand Dwelling house/Apartment”, accounting for approximately 82.2% of transactions, while “New Dwelling house/Apartment” represents approximately 17.8%. This indicates that the majority of property transactions involve previously owned properties rather than newly constructed homes. The distribution of this feature can be seen in Appendix B.5.

An additional data quality issue was identified within the *Description of Property* feature. Some observations record the category “Second-Hand Dwelling house/Apartment” using its Irish translation “Teach/Árasán Cónaithe Athláimhe”. Although these values represent the same property type, they appear as separate categorical levels in the dataset. This introduces unnecessary duplication in the category labels and slightly inflates the cardinality of the feature. In later preprocessing steps, these labels will be standardised to ensure consistent representation of property types.

The **Property Size Description** feature exhibits a very high proportion of missing values, with approximately 94.82% of observations lacking size information. Among the non-missing values, the most common category is “greater than or equal to 38 sq metres and less than 125 sq metres”, representing approximately 3.87% of the dataset. Due to the extremely high missingness, this feature may have limited usefulness for predictive modelling unless an appropriate strategy for handling missing values is implemented. The distribution of the recorded values for this feature is presented in Appendix B.6.

The **Eircode** feature can be considered a categorical variable; however, it behaves closer to an identifier due to its extremely high cardinality (16,840 distinct values) and sparse distribution. Additionally, approximately 68.71% of observations have missing Eircodes. The most frequent Eircode appears only three times, indicating that the feature provides very limited generalisable information. This combination of high cardinality and substantial missingness limits its usefulness for predictive modelling.
An additional issue is the uneven temporal and geographic distribution of Eircodes. Earlier years in the dataset contain very few recorded Eircodes, while later years show increased coverage. Furthermore, a disproportionate number of recorded Eircodes are associated with Limerick, despite Dublin being the most frequent county overall. This inconsistency suggests that Eircode availability is influenced by data collection practices rather than true geographic distribution, further limiting its

reliability as a predictive feature. A visualisation of the most common Eircode Routing Keys can be found in Appendix B.7.

Overall, the categorical feature analysis reveals several important data quality issues, including high cardinality in the Eircode feature and substantial missingness in the Property Size Description feature. Additionally, several categorical variables exhibit highly imbalanced distributions, which may influence their usefulness in predictive modelling.

Other Data Types

In addition to the continuous and categorical features described previously, the dataset contains two additional variables with specialised data types: *Date of Sale* and *Address*. These features do not fit naturally into the standard continuous or categorical data quality tables, therefore they are assessed separately in terms of structure, completeness and potential usefulness for modelling.

The **Date of Sale** feature represents the date on which each property transaction occurred. The values were successfully parsed into a datetime format using the dd/mm/yyyy structure. The dataset covers transactions from 2016 to 2024, providing approximately nine years of property sales data. No missing values were observed for this feature, indicating complete temporal information for all observations.
Temporal variables such as this are useful for capturing trends in property prices over time. In later stages of the analysis, the date feature may be decomposed into additional features such as sale year or sale month in order to capture temporal patterns in the housing market.

The **Address** feature is a free-text field containing the full postal address of each property transaction. As expected, the feature exhibits extremely high cardinality, as most addresses appear only once in the dataset. However, analysis revealed 699 duplicate addresses, indicating that some properties were sold multiple times during the period covered by the dataset. These duplicates represent legitimate repeated property transactions rather than data entry errors.

The address strings are not uniformly formatted and may contain house numbers, apartment identifiers, estate names, towns or abbreviated county names. This lack of standardisation makes the feature difficult to use directly in predictive models.

Despite these limitations, the address field may still contain useful geographic information. In later stages of the analysis, elements of the address could potentially be used to derive additional features such as locality names, Dublin postal districts, or indicators of apartment properties.

Structural Issues & Initial Cleaning Actions

Overall, the dataset is relatively complete and well structured. Most rows contain valid values for the majority of features, particularly the key variables such as *Price*, *County*, *Description of Property* and *Date of Sale*. However, several structural and statistical issues were identified during the data exploration process which require attention before predictive modelling can be performed.

Some features exhibit significant levels of missing data. In particular, the *Property Size Description* feature contains very few recorded values, with the vast majority of entries missing. Similarly, the *Eircode* feature contains a large proportion of missing values. These issues limit the usefulness of these variables and will require further consideration in the Data Quality Plan.

Certain features, particularly *Eircode* and *Address*, exhibit extremely high cardinality, with most values appearing only once. High-cardinality categorical variables can be difficult to use directly in predictive models because there are insufficient observations per category to learn meaningful patterns.

The target variable *Price* exhibits a heavily right-skewed distribution, with a small number of extremely high-value transactions producing a long upper tail. These extreme values stretch the distribution and may influence model performance. However, such values likely represent legitimate high-value property transactions rather than data entry errors.

Several formatting inconsistencies were identified in the raw data. For example, the *Price* feature included currency symbols (€) and comma separators, which prevented it from being interpreted as a numeric variable. Similarly, the *Date of Sale* feature was initially stored as a string rather than a datetime object.

Minor inconsistencies were also identified in categorical labels. For example, the category "Second-Hand Dwelling house/Apartment" also appears in its Irish translation "Teach/Árasán Cónaithe Athláimhe". Although both values represent the same property type, they appear as separate categories in the dataset.

No duplicate columns or constant columns were identified in the dataset, indicating that all features provide at least some level of variability.

Initial Cleaning Steps Applied

Several initial cleaning steps were applied to prepare the dataset for further analysis:
The main data quality concerns identified during the exploration phase are summarised below:

- Currency formatting (€, commas) was removed from the *Price* feature and the column was converted to a numeric data type.
- The *Date of Sale* column was converted to a datetime format.
- Basic validation checks were performed to ensure the dataset contained the expected number of rows and columns.
- Duplicate rows were removed from the dataset.
- Cleaned file saved as: ppr-group-25200353-train-cleaned.csv

Summary of Key Data Quality Concerns

The main data quality concerns identified during the exploration phase are summarised below:

- Extreme outliers in Price
- 94.8% missing values in the *Property Size Description* feature.
- 68.7% missing values in the *Eircode* feature.
- Extremely high cardinality in geographic features such as *Eircode* and *Address*.
- A heavily right-skewed distribution of the *Price* target variable.
- Minor inconsistencies in categorical labels.

Next Steps

While several structural issues were addressed during the initial cleaning phase, a number of data quality challenges remain. These include handling missing values, addressing high-cardinality features, and determining appropriate strategies for managing skewness in the target variable. These issues will be formally addressed in the Data Quality Plan, where specific solutions and preprocessing steps will be proposed and implemented.

Appendices

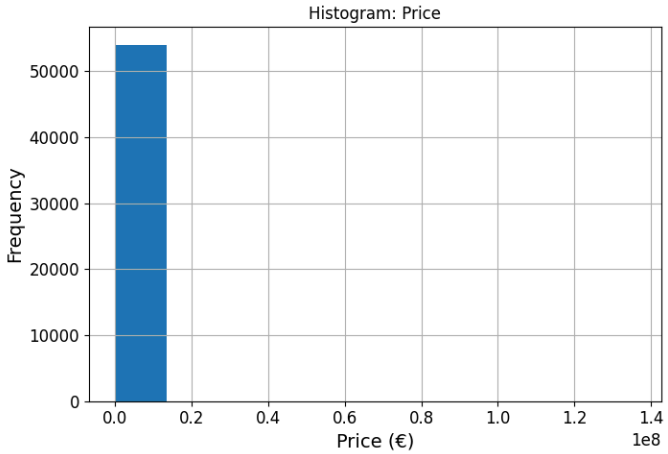
The following appendices provide supporting visualisations and summary tables referenced throughout the Data Quality Report. High-resolution versions are available in the accompanying analysis notebook.

Appendix A - Continuous Features

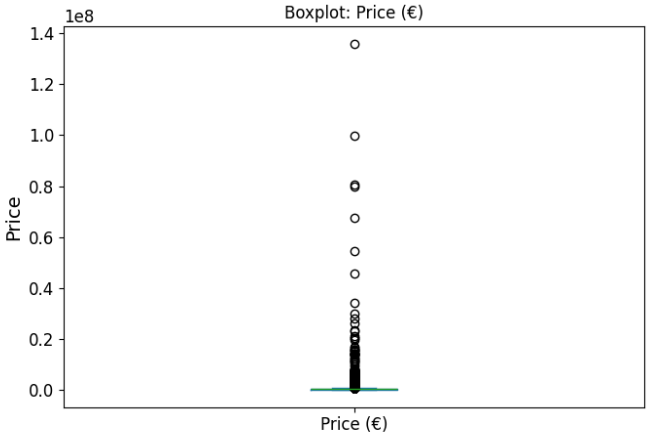
Appendix A.1 - Data Quality Report Table for Continuous Feature

	Feature	Count	Missing %	Cardinality	Minimum	1st Quartile	Mean	Median	3rd Quartile	Maximum	Standard Dev.
0	Price (€)	53992	0.0	6788	5260.0	158590.33	327045.85816	252000.0	365000.0	1.358598e+08	1.108129e+06

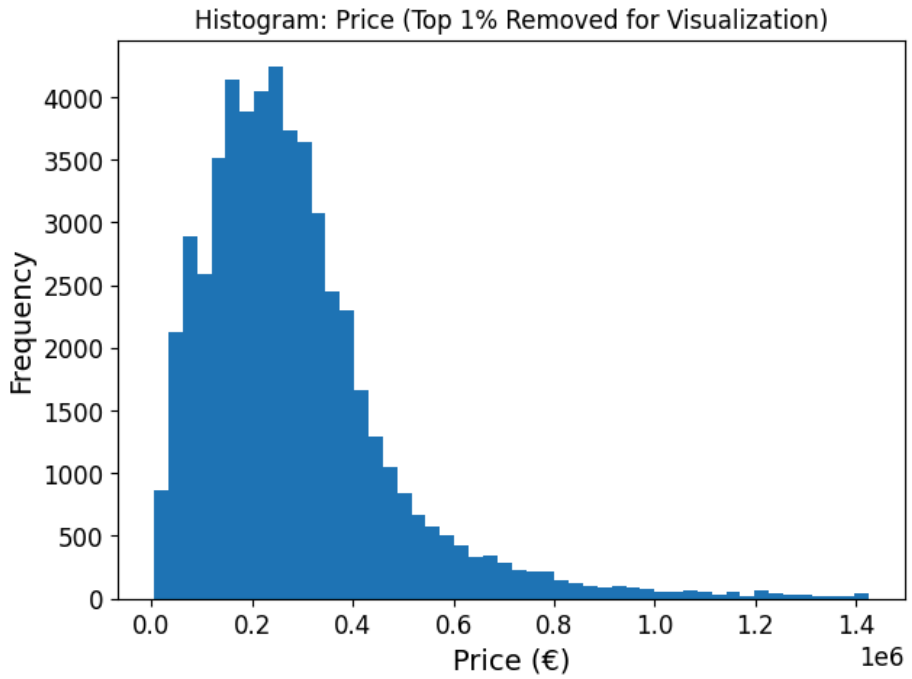
Appendix A.2 - Price Histogram on Raw Price Values



Appendix A.3 - Boxplot to View Price Outliers



Appendix A.4 - Histogram of Top 1% of Values Removed to Show Without Extreme Right Skew

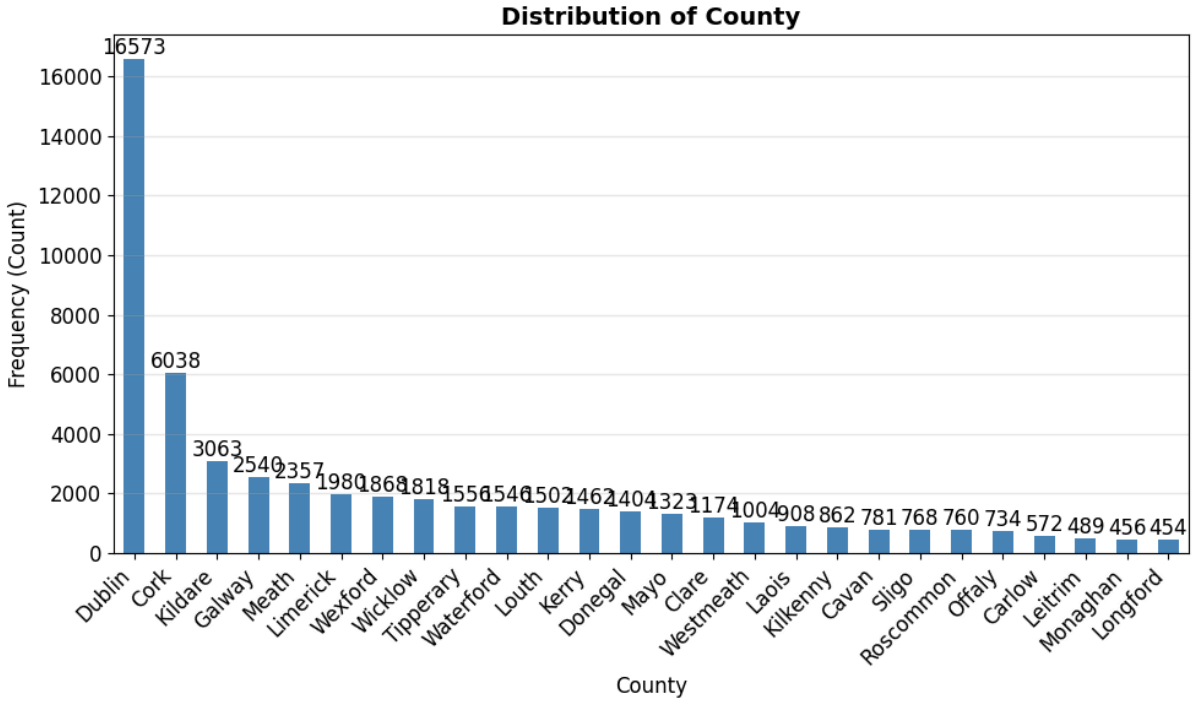


Appendix B - Categorical Features

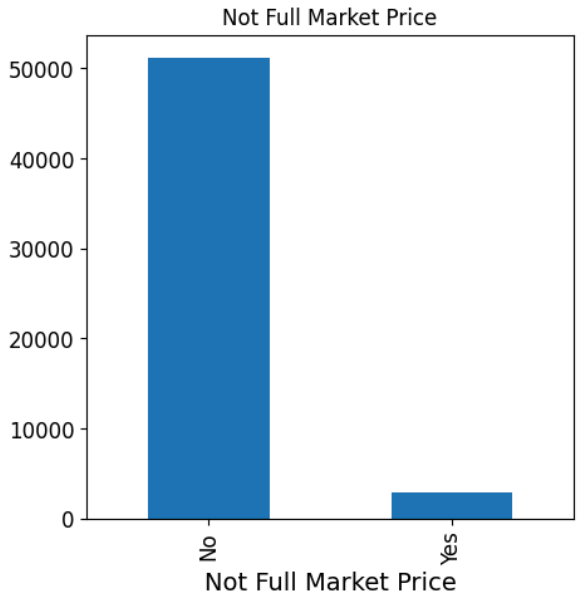
Appendix B.1 - Data Quality Report Table For Categorical Features

	Feature	Count	Missing %	Cardinality	Mode	Mode Freq.	Mode %	2nd Mode	2nd Mode Freq.	2nd Mode %
0	County	53992	0.000000	26	Dublin	16573	30.695288	Cork	6038	11.183138
1	Eircode	53992	68.712031	16840	D24W9NN	3	0.005556	D08X5T9	3	0.005556
2	Not Full Market Price	53992	0.000000	2	No	51143	94.723292	Yes	2849	5.276708
3	VAT Exclusive	53992	0.000000	2	No	44523	82.462217	Yes	9469	17.537783
4	Description of Property	53992	0.000000	3	Second-Hand Dwelling house /Apartment	44359	82.158468	New Dwelling house /Apartment	9630	17.835976
5	Property Size Description	53992	94.815899	4	greater than or equal to 38 sq metres and less...	2088	3.867240	greater than or equal to 125 sq metres	493	0.913098

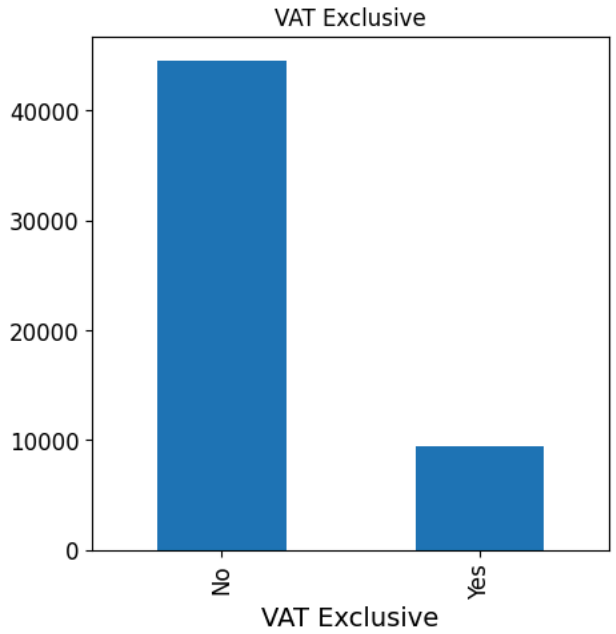
Appendix B.2 - Histogram of County Distribution



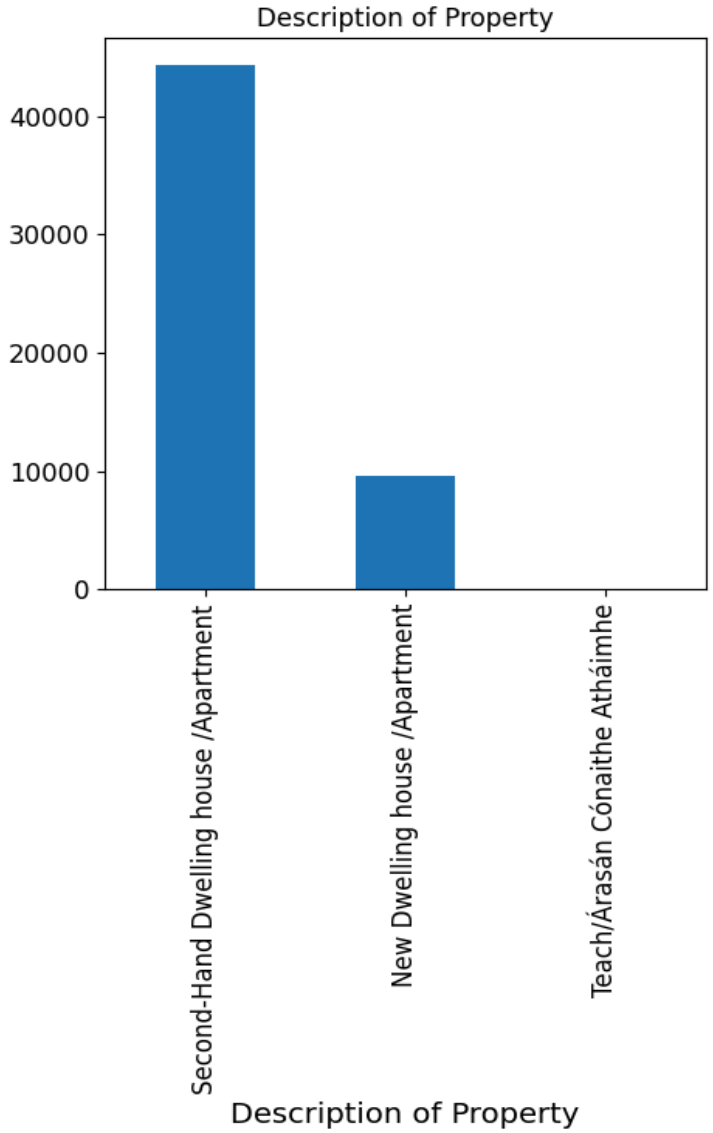
Appendix B.3 - Histogram showing Not Full Market Price Distribution



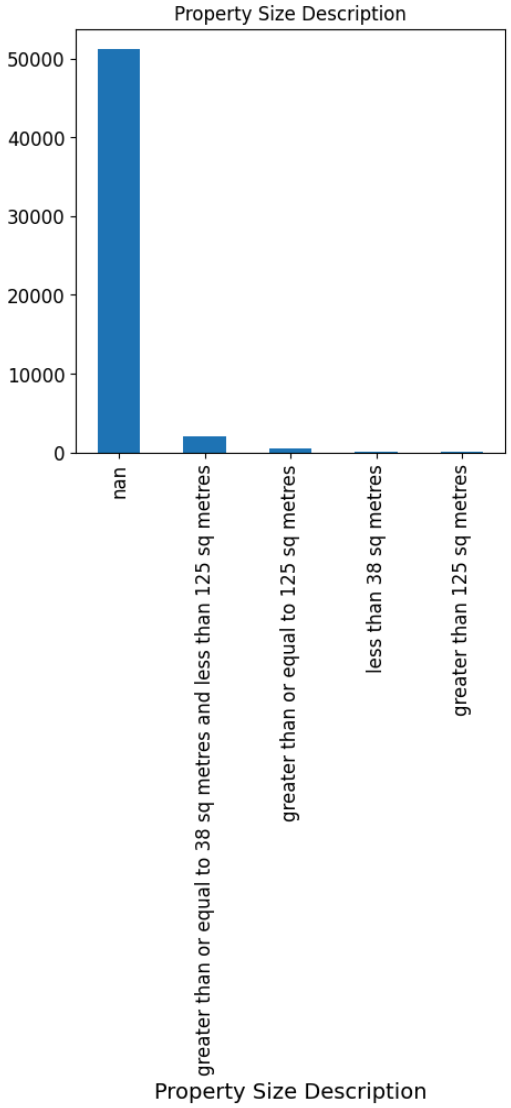
Appendix B.4 - Histogram of VAT Exclusive Distribution



Appendix B.5 - Histogram of Description of Property Distribution



Appendix B.6 - Histogram of Property Size Description



Appendix B.7 - Top 35 Most Common Eircodes in Data

